Project Proposal: Phase 1

Consent-Based Smart Receptionist Kiosk

CMAI 480 Computer Vision · Dr. Khalil Ahmad Yousef

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1. What it is

A visitor who has an appointment walks up to a kiosk instead of waiting for the receptionist. Before the camera turns on, the screen asks the visitor for permission to use their face. After they agree, the kiosk finds the face, checks that it is a real person and not a photo on a phone screen, turns the face into a set of numbers, and compares those numbers with the list of people expected that day. If it finds a match, the visitor is checked in and the receptionist is told. The system has four parts: face detection, feature extraction to make the numbers, liveness checking, and a MATLAB App Designer screen for the visitor and the receptionist.

2. Why it is worth doing

Right now the receptionist checks every visitor's name against a list by hand, one person at a time, while the others wait. This is slow when a few people arrive together, it is the same job repeated all day, and the visitor has to say their name out loud in a full waiting room. We want checking in to take a few seconds, and we want it to happen without the visitor losing their privacy.

3. Our ideas on top of the title

- Two equal ways to check in. Face recognition cannot work for some visitors at all. They may wear a face covering, they may wear a medical mask, or they may simply say no. So we build the second way of checking in as an equal path, not as a backup. We will time both ways and show the results side by side.
- We only match against today's list. We never search a big database. We only compare with the people booked for that day, which is usually 10 to 50 people. This is why a simple MATLAB method is enough here. We will show this by drawing a graph of the accuracy as the list grows from 5 people to 50 people.
- We compare our own code with a ready made model. We write the PCA (Eigenfaces) method ourselves from the equations, then we add a trained model as a second method and show the difference between the two on our own faces. Our own method becomes the result that everything else has to beat.

4. How we will know it worked

- Accuracy: how well the system recognises people in our own test set, compared with our own PCA method as the score to beat.
- Liveness: we make our own fake samples (a printed photo, a phone screen and a laptop screen) and report how often each one fools the system, so we can see which attack is the hardest to stop instead of giving a single average.
- Speed: how long a full check in takes, using the face way and using the other way.
- Failures: which conditions break the system, such as strong backlight, glasses, a mask, or turning the head too far.

5. What we are not doing

- No group check in. One visitor at a time.
- No cloud, no phone app, and no link to a real booking system.
- No doors, gates, badge printing or payments.

6. Plan

Phase 1 (weeks 1 to 3): requirements, consent process, and collecting the first dataset.

Phase 2 (weeks 4 to 5): reading related work, cleaning the images, and face alignment.

Phase 3 (weeks 6 to 9): our own PCA method, a better method, and the liveness classifier.

Phase 4 (weeks 10 to 12): live camera, the App Designer screen, experiments, and failure analysis.

Phase 5 (week 13): final report and code documentation.

Phase 6 (weeks 14 to 15): presentation, live demo, and the individual oral exam.

7. Question related to an idea we want to use please Doctor answer it:

- Can we use a ready made face recognition model, and which ones are allowed? We would use it together with our own PCA method, not instead of it. It would save us a lot of time on building and tuning the matching step, and it should give better accuracy than our own method alone. We would load it inside

MATLAB with the Deep Learning Toolbox, so we would not use Python or any program outside MATLAB, and we would still write the PCA method ourselves from the equations as the brief asks.